Release Notes

Version 5.0.0 Pre-Release.1 - June 9, 2026

General Information

Following the recent **hardware upgrades**, we've opted to stop using GPIO pins to control the platforms. Instead, we will rely on a dedicated micro-controller. This means that a new software revision is required, and that this revision is technically **major** as all previous versions will no longer be supported. For now, this pre-release only contains the necessary changes to make the platforms functional in experiment, plus a couple small bug fixes. Expect more significant changes down the line as we try to move towards more version-control friendly GUI. **Users should only really upgrade to this pre-release version if they intend to conduct experiments imminently. All other users should wait until the full release later this summer.**

Initializer Changes

N/A

GUI Changes

- **Change 1:** Updated all hardware interface code to enable control through the GUI using the new micro-controller. This includes turning the pucks off and on, as well as checking all the thrusters.
- **Change 2:** Updated the default experiment initial conditions so that it puts all platforms through their paces equally.
- **Change 3:** Updated the mass properties and thruster locations to reflect the hardware changes.

Simulink Changes

- **Change 1:** Removed a commented out block from the data saving subsystem which is not being used.
- **Change 2:** Removed the accelerometer/gyroscope sensor susbsystem as the sensors have been removed in favor of using those onboard the LiDAR. **This update cannot currently access these measurements by default**.
- **Change 3:** Replaced the UDP send block for the thruster duty cycle commands with the serial send block. Thruster duty cycle commands are now sent over serial to the standalone micro-controller.
- **Change 4:** Updated the default experiment so that it puts all platforms through their paces equally.

Bug Fixes

- **Bug 1:** Fixed a bug in the saveExperimentData GUI sub-function, which could sometimes cause saving experiment data to fail when the data contains NaNs
- **Bug 2:** Fixed a bug in the ToggleRealTimeTelemetrySwitchValueChanged GUI sub-function, which could result in an error when live streamed data contains NaNs. If received data contains NaNs, then the last received data is retained.

Ongoing Issues

- **Issue 1:** Due to a problem with the control panel on the chaser spacecraft, the emergency stop switch does not work with the chaser. This will be fixed in the coming weeks and will not required a new software revision.

Acknowledgments

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- **Jeremy Peters**: Thank you for all your volunteered time with the hardware revisions. This would not have been possible to complete in so little time without your help. Thanks as well for working out the new communication code for the micro-controller!
- **Sarah Muzika**: Thank you for all your hard work on soldering together the new components. Hopefully, these new hardware parts will prove to be more reliable when compared to what came before!
- **Aaron vandenEnden**: Thank you for helping out with some of the CAD files we needed for the new hardware!